

Design of Bifunctional Fentanyl Analogs: Dual-Targeting the Mu-Opioid and Nociceptin Receptors for Safer Analgesia

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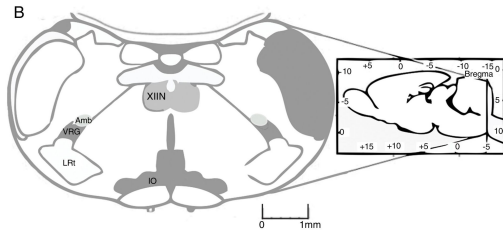
Motivation

Fentanyl is the primary driver of fatal opioid overdose. Between 2019 and 2023, Oregon experienced a 1,500% increase in fentanyl-related overdose deaths, the highest in the United States. [1]

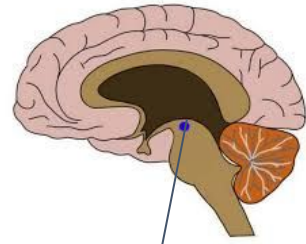
Fentanyl Overview

Low-cost synthetic opioid that is 100x stronger than heroin & morphine. Binds to the Mu-Opioid Receptor (MOR) in the Central Nervous System via the **Asp147** residue. [2]

Fentanyl Challenges



PreBotzinger Complex [1]



Ventral Tegmental Area [2]

- Binding in the PreBötzing Complex induces respiratory failure [2].
- Binding in the Ventral Tegmental Area (VTA) promotes dopamine release and addictive potential. [3].

Existing Alternatives

Morphine:

- Long history, widely available
- Slower onset, less potency than fentanyl
- Maintains risks of respiratory depression and dependence

Acetaminophen/Ibuprofen:

- Very safe, Over-the-Counter Accessible
- Significantly weaker analgesic effect

Existing Treatments

Naloxone (Narcan):

- **Reverses** Opioid overdose by **blocking** off Mu-Opioid Receptors [4]
- **Symptomatic** measure - Cannot change inherent risk of opioids
- **No effect** on addiction potential - Does not inhibit dopamine release

Problem Statement: Fentanyl carries heavy side-effects, while no robust alternatives or treatments exist

Problem Approach

General Drawbacks ——— Optimal Solution

PROBLEM

MOR Targeting Compounds carry heavy side-effects

SOLUTION

Nociceptin/Orphanin FQ receptor (NOP)

Pharmacological breakthroughs have shown NOP regulates the same systems that MOR overstimulate. Dual-activation reduces side-effects [5].

PROBLEM

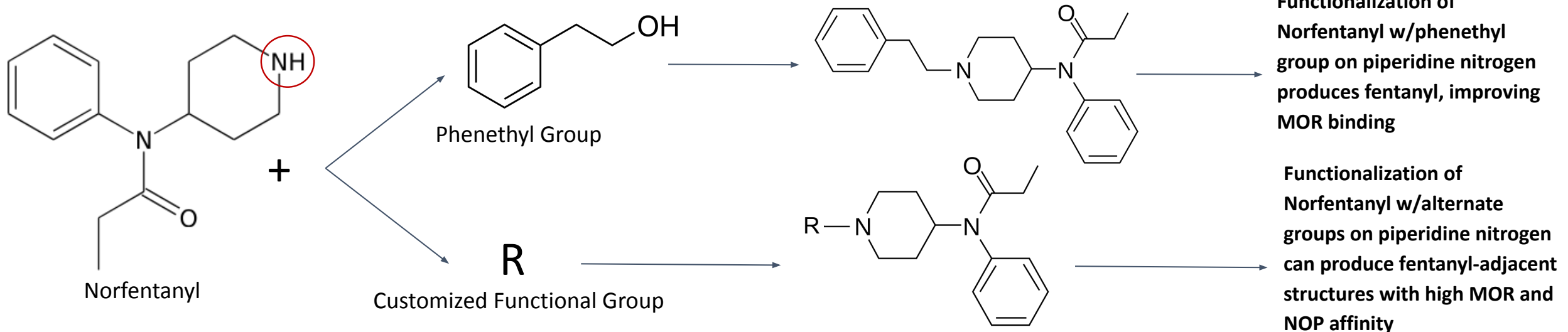
Designing entirely new compounds is slow and high-risk

SOLUTION

Modification of Fentanyl—Preserves validated, optimal scaffold

Keeping the fentanyl structure can preserve its pharmacokinetics and potency while selectively altering interactions to boost NOP activation

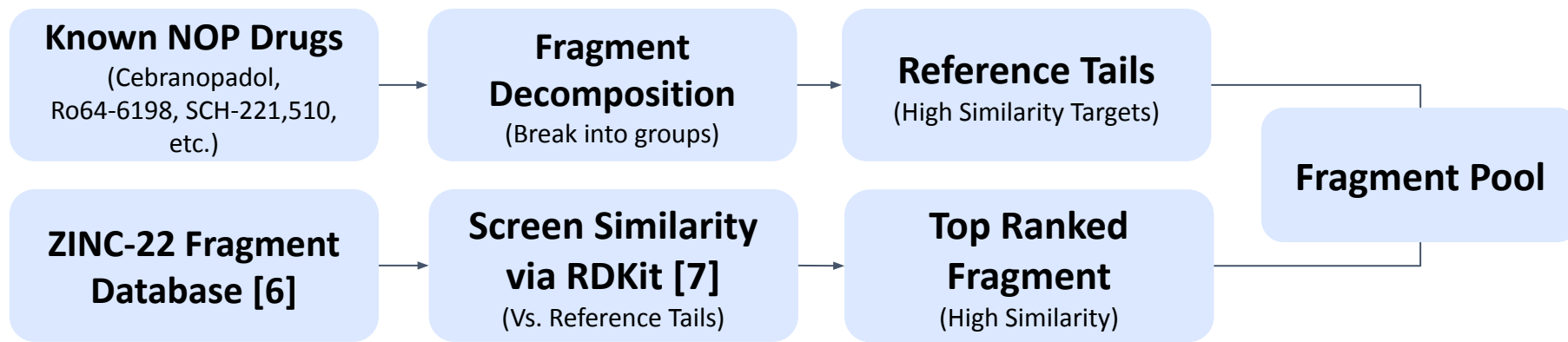
Proposed Solution



Summary: Modification of Norfentanyl Scaffold can produce Fentanyl-like Structures with higher NOP Affinity, Reducing side-effects

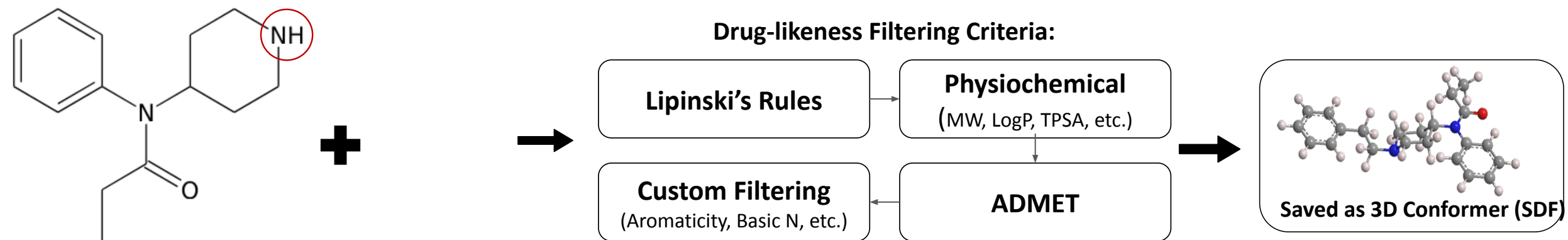
Functional Group Identification and Analog Preparation

Fragment Building



Explanation: By analyzing the commonalities between known NOP agonists, we can recognize and leverage functional groups likely to contribute to NOP activity

In-Silico Preparation



Summary: Intensive Chemical filtering via Python RDKit identifies 57 Norfentanyl-derived compounds that can be analyzed for NOP binding

Methods: Target Protein Preparation



PDB ID 5C1M: MOR/BU72 Complex [1]

ChimeraX

1. Binding Coordinates

Binding Coordinates are identified (X: 1.29, Y: 16.45, Z: -59.11)

3. Remove Extra-Residues

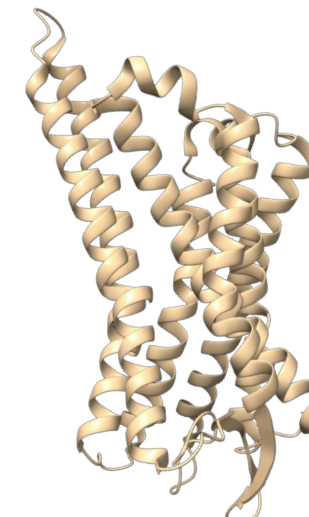
All non-standard residues (ions, H₂O, ligands) that stop binding are removed

2. Remove Bound Drug

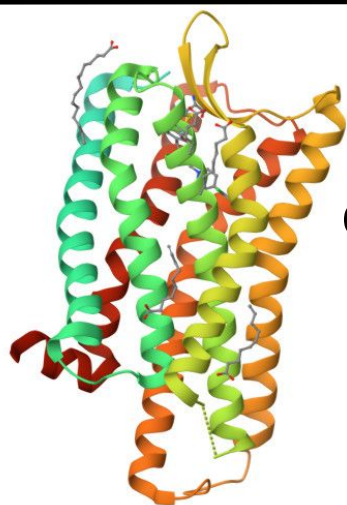
BU72 is deleted from structure to isolate MOR Protein

4. Add Hydrogen (H) atoms

Polar Hydrogen bonds are added to ensure proper protein structure



Pristine MOR Protein



PDB ID 5DHH: NOP/SB-612111 Complex [1]

ChimeraX

1. Binding Coordinates

Binding Coordinates are identified (X: -9.23, Y: 29.04, Z: -2.71)

3. Remove Extra-Residues

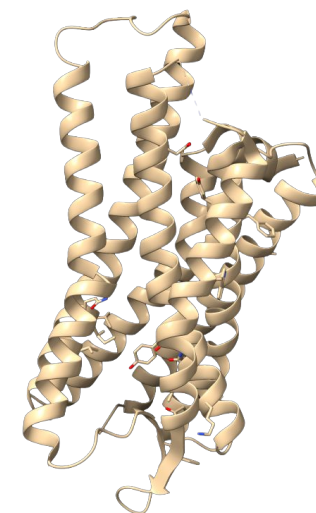
All non-standard residues (ions, H₂O, ligands) that stop binding are removed

2. Remove Bound Drug

SB-612111 is deleted from structure to isolate MOR Protein

4. Add Hydrogen (H) atoms

Polar Hydrogen bonds are added to ensure proper protein structure



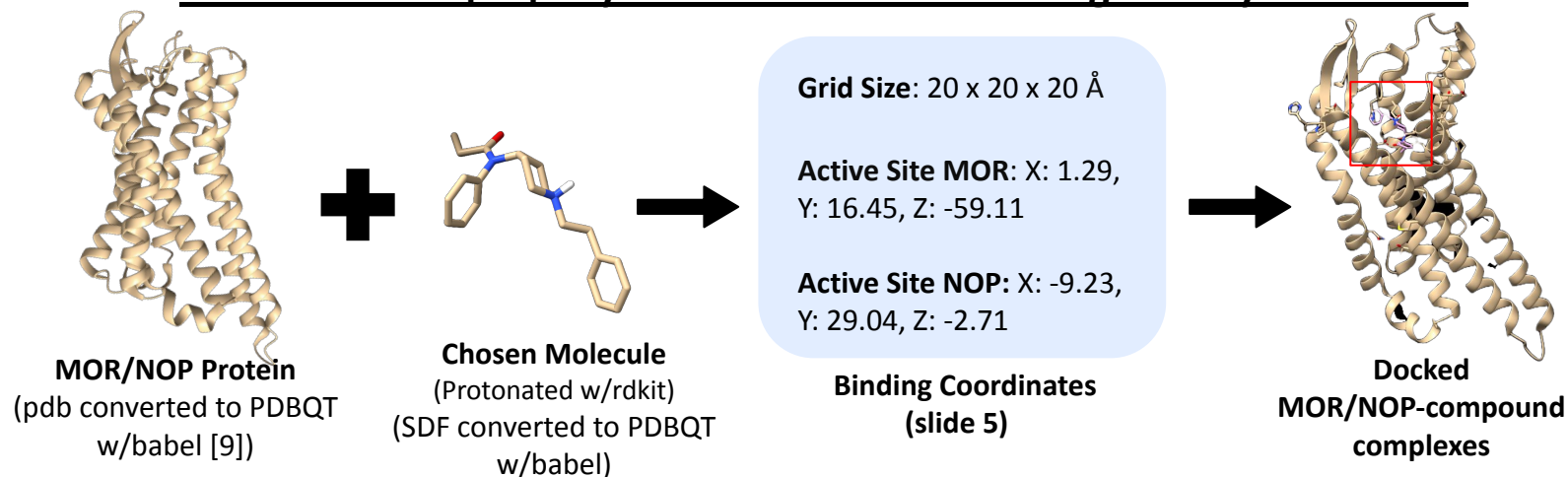
Pristine NOP Protein

Summary: MOR & NOP proteins are prepared by cleaning 2 protein + drug complexes in ChimeraX [8], and saving for molecular docking

Methods: Molecular Docking + Dynamics

Simulation Protocol

AutoDock Vina [10] - Python Extension for Binding Affinity Simulation



Vina Scoring Function

Vina calculates the binding affinity of two molecules by estimating gibbs free energy, ΔG , given by:

$$\Delta G = \Delta H - T\Delta S$$

(H = Enthalpy, T = Temperature, S = Entropy)

Negative scores indicate spontaneous reaction

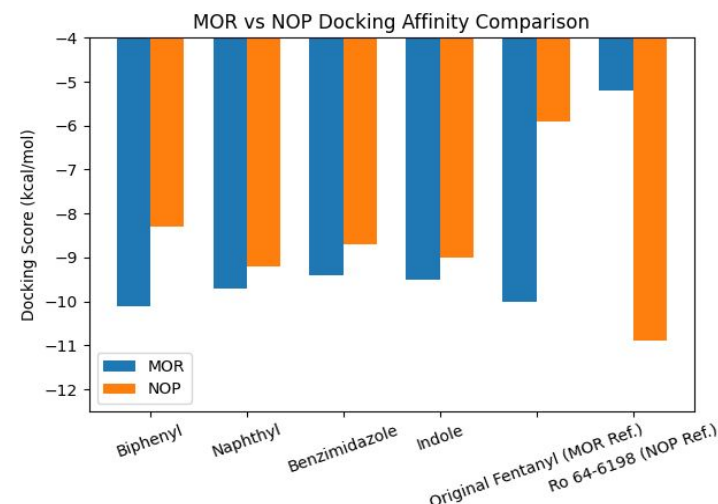
(High binding affinity)

57 Compounds
filtered for MOR & NOP $\Delta G < -9.0$ kcal/mol, and saved for **molecular dynamics (MD)**.

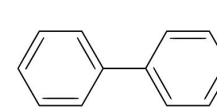
4 resultant compounds.
Converted to .xml files. **MD performed** w/OpenMM [11].

Top Docking Candidates

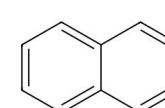
MOR	NOP
Biphenyl: -10.1 kcal/mol	-8.3 kcal/mol
Naphthyl: -9.8 kcal/mol	-9.2 kcal/mol
Benzimidazole: -9.4 kcal/mol	-8.7 kcal/mol
Indole: -9.5 kcal/mol	-9.0 kcal/mol



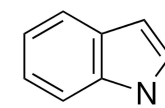
- Chosen Functional groups show **similar MOR/NOP affinity** to established agonists



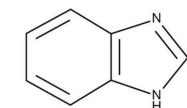
Biphenyl



Naphthyl



Indole



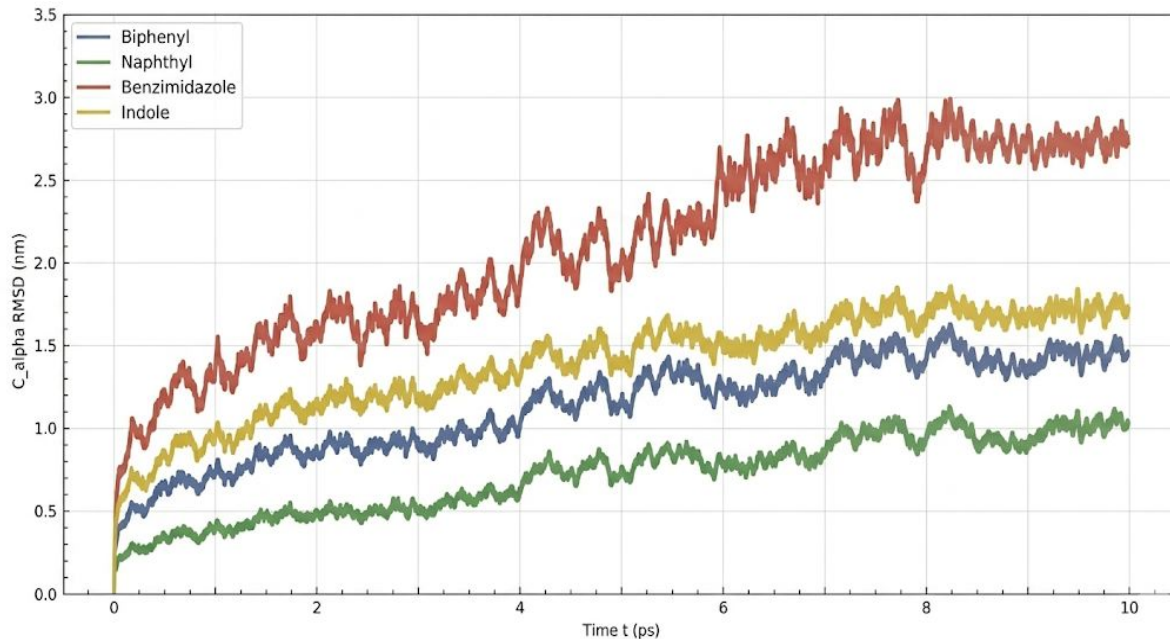
Benzimidazole

Summary: AutoDock Vina was used to screen preliminary compounds. Compounds with binding affinity < -9.0 kcal/mol were saved for molecular dynamics simulations.

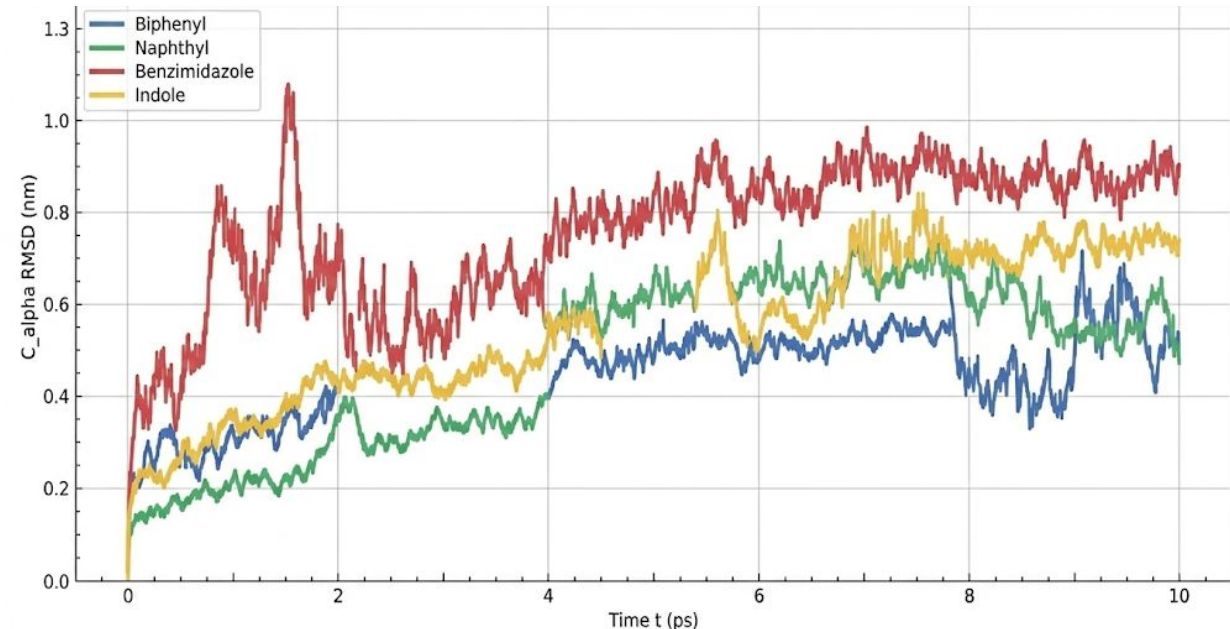
Results: Molecular Dynamics (MD)

MD is a technique for simulating a live protein-ligand interaction. Helps identify most stable scaffold(s).

Mu-Opioid Receptor



Nociceptin Receptor

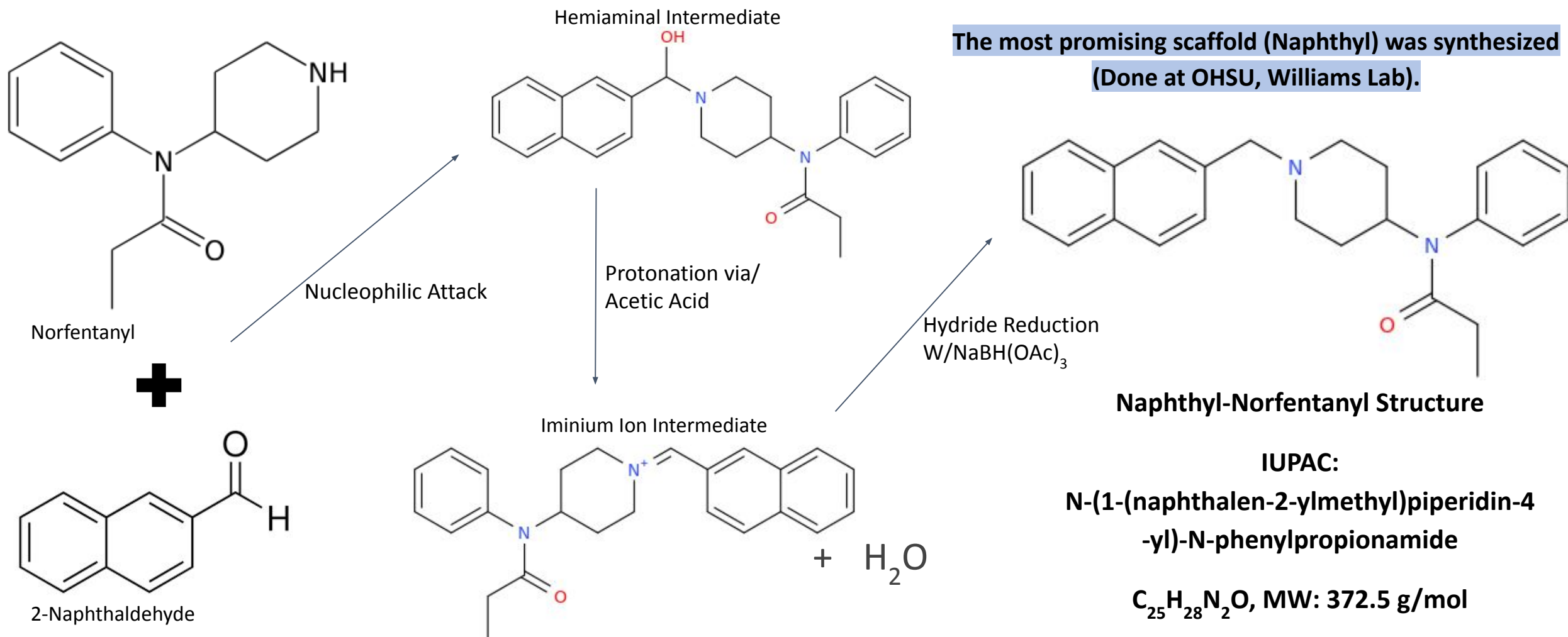


All molecules showed Root Mean Square Deviation (RMSD) of <3.0 Angstrom. Lower values Indicates high binding stability. Naphthyl modified scaffold displayed lowest RMSD across both MOR and NOP. Naphthyl-Scaffold shows highest promise for in-vitro validation.

Summary: All compounds show stable molecular dynamics results and interact with receptors with correct residues (ASP147 & ASP130), Naphthyl-based scaffold particularly promising

Methods: Analog Synthesis

One-pot Functionalization of Norfentanyl W/Naphthyl Group

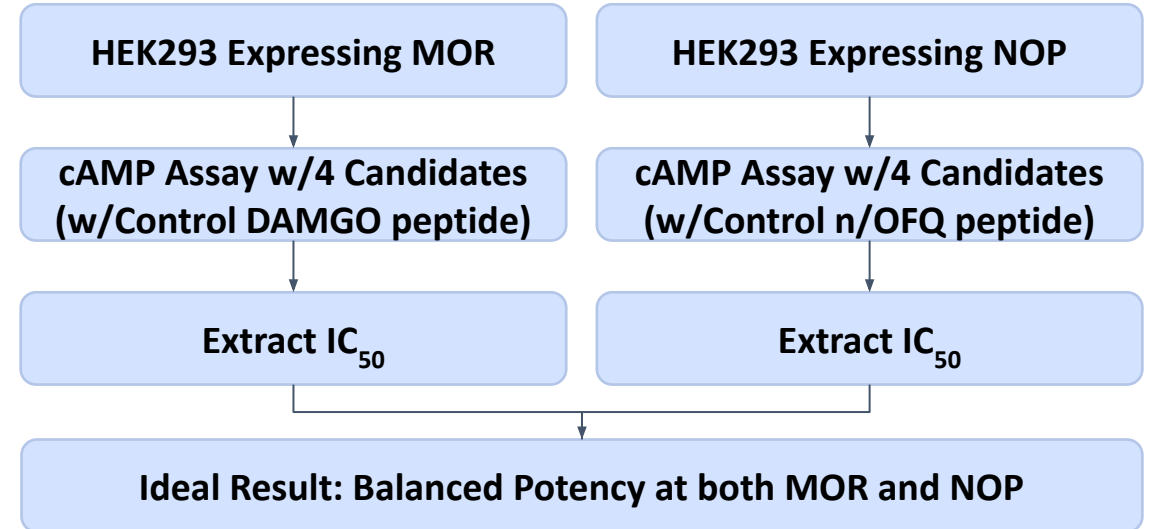
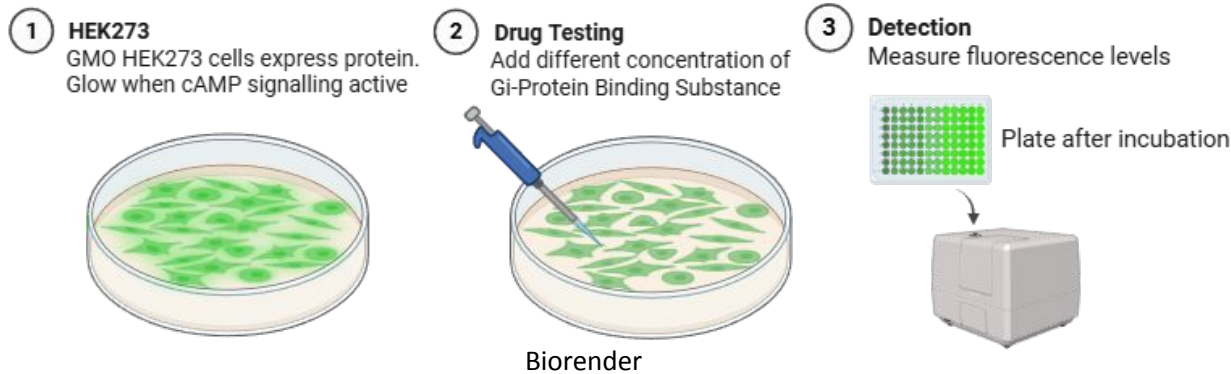


Summary: Naphthyl-Norfentanyl Compound Synthesized through Reductive Amination reaction using 2-Naphthaldehyde

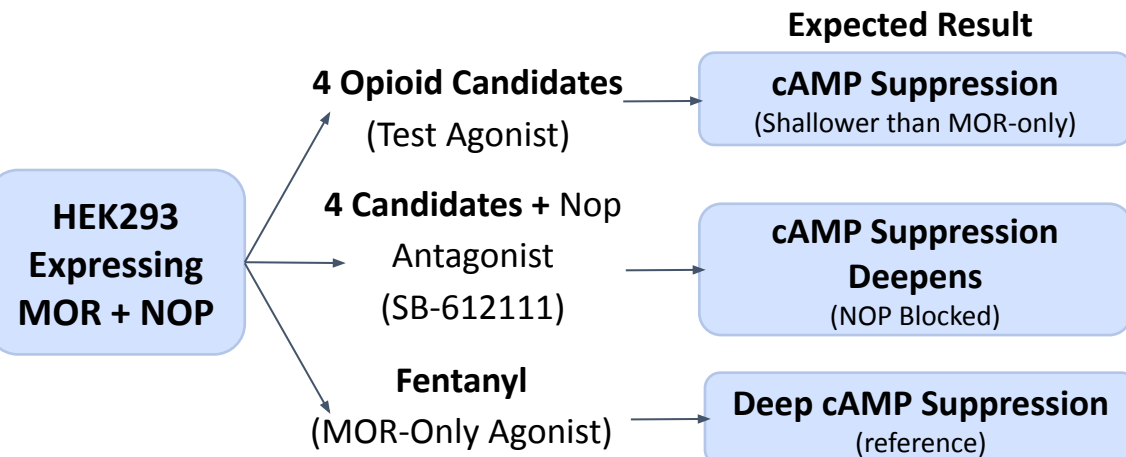
Methods: In-Vitro Compound Validation

Individual cAMP Assays

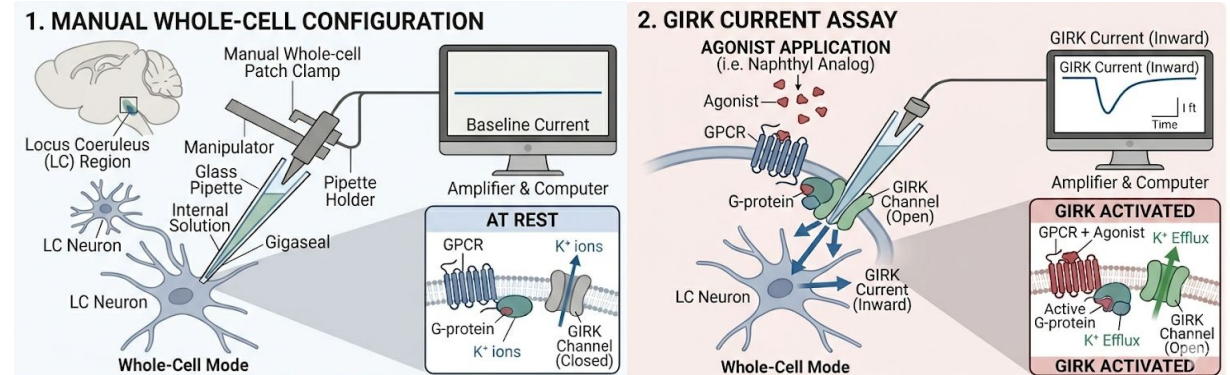
cAMP Assay: a method that measures a cell's signaling activity; cAMP is high (and can produce a fluorescent signal) when Gi-coupled receptors like MOR or NOP are inactive, and drops (dims) when they're activated.



Dual-Activation cAMP Assay



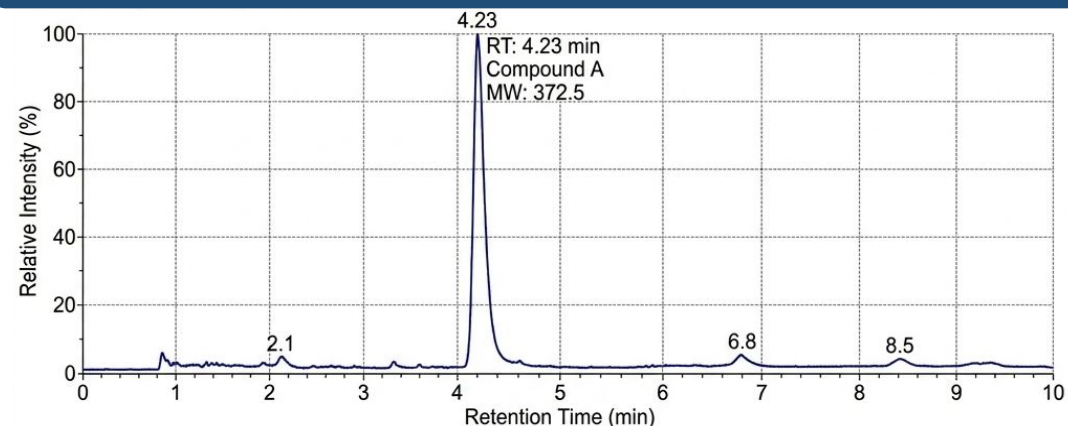
Brain Slice GIRK Assay



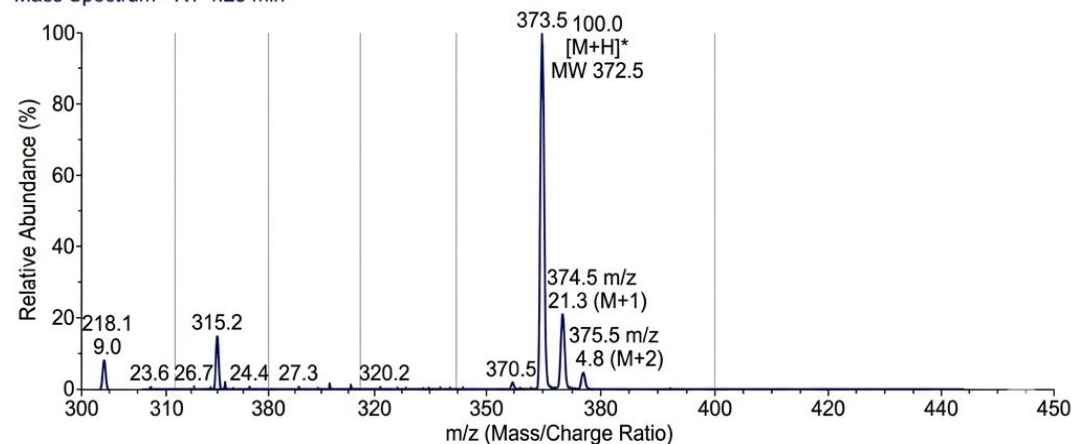
GIRK channels serve as a readout in electrophysiology assays to measure how effectively a drug activates a specific receptor signaling pathway.

Results: Synthesis Characterization

Liquid Chromatography-Mass-Spectrometry



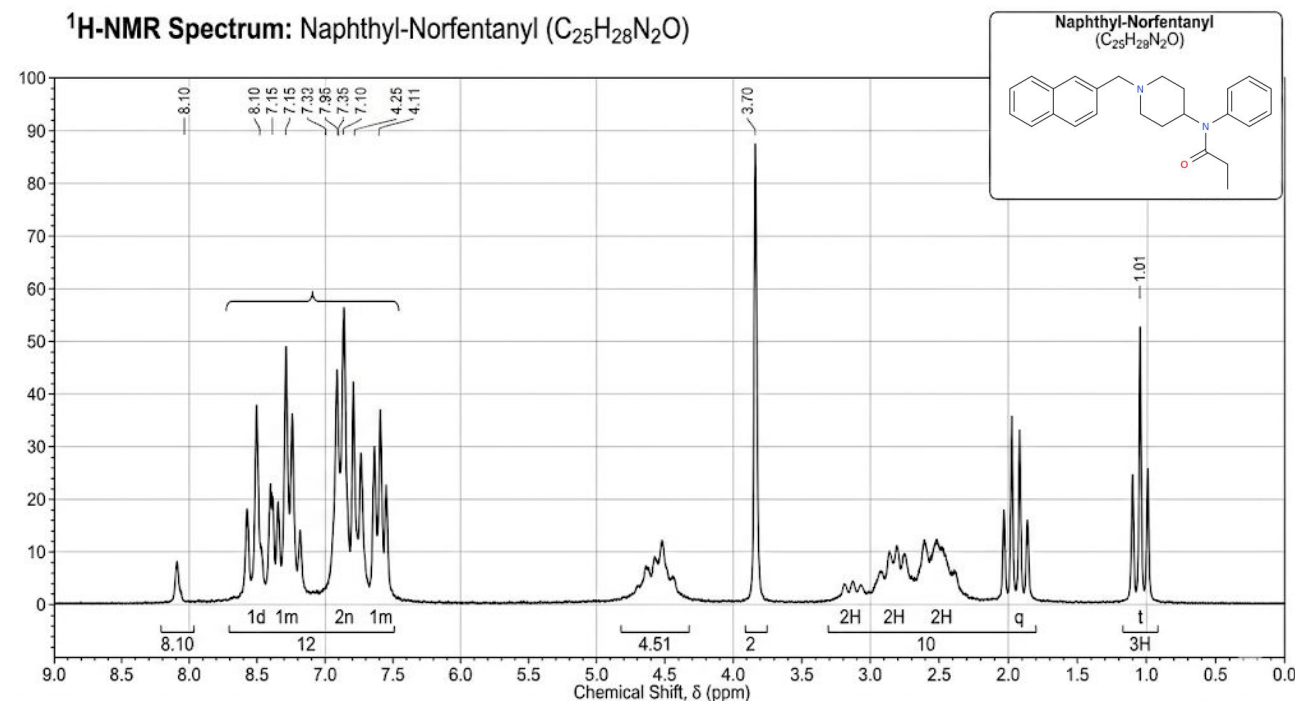
Mass Spectrum - RT 4.23 min



- LC Shows correct intensity at 4.23 minutes
- MS Shows correct MW of ~372.5 g/mol

Proton Nuclear Magnetic Resonance (H-NMR)

¹H-NMR Spectrum: Naphthyl-Norfentanyl (C₂₅H₂₈N₂O)

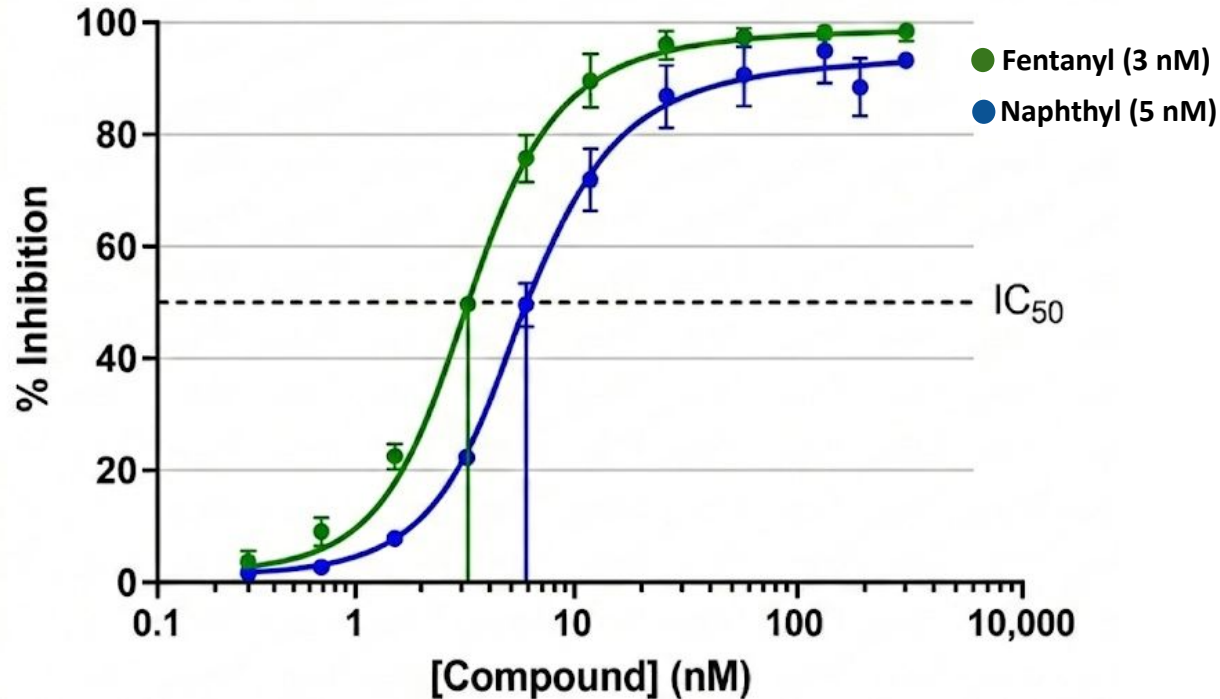


- H-NMR Shows **12 aromatic protons** (naphthyl + phenyl rings), a **2H singlet** at 3.70 ppm marking a **methylene group**, and 2H quartet and 3H triplet between 1.0 and 2.0 ppm showing **propionyl group**

Summary: Successful synthesis of Naphthyl-Norfentanyl confirmed w/LC-MS & H-NMR (94.8% purity)

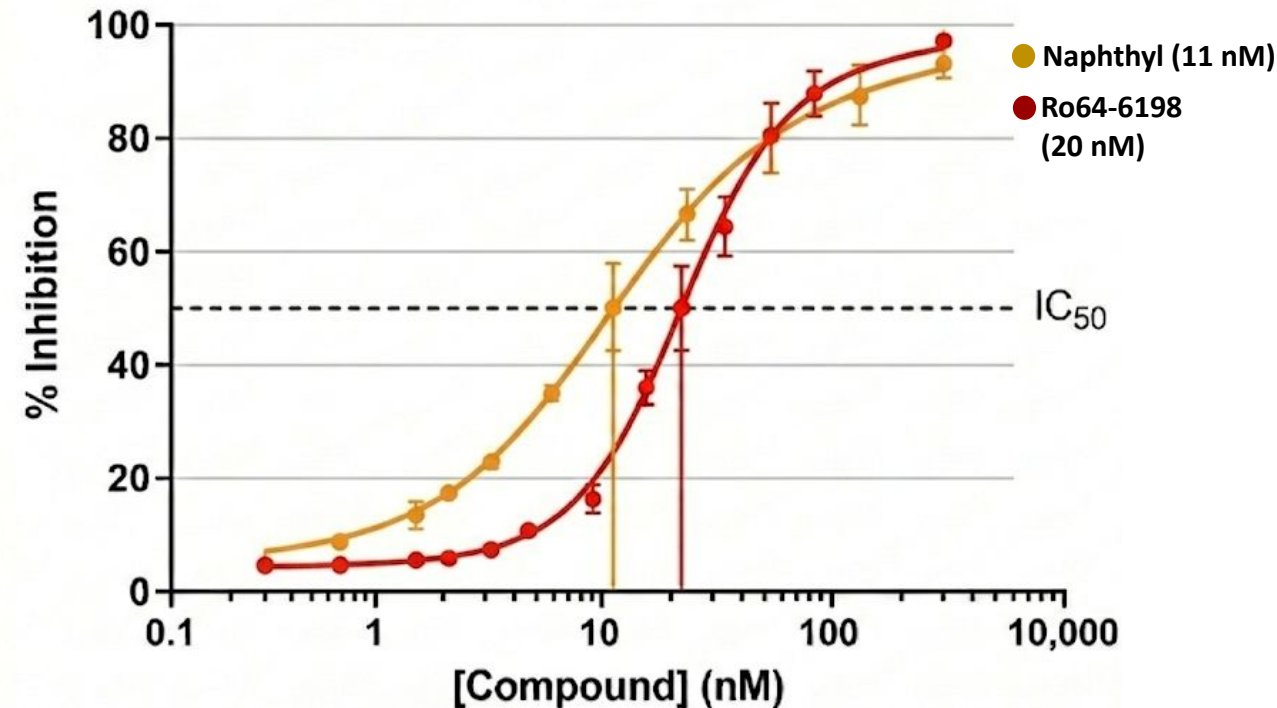
Results: Individual In-Vitro cAMP Signalling

IC50 Values - MOR



The Naphthyl-modified norfentanyl scaffold showed an half-maximal inhibitory value (IC50) of 5 nanomoles. This is comparable to Fentanyls IC50 of 3 nanomoles, showing similar MOR activation potency

IC50 Values - NOP

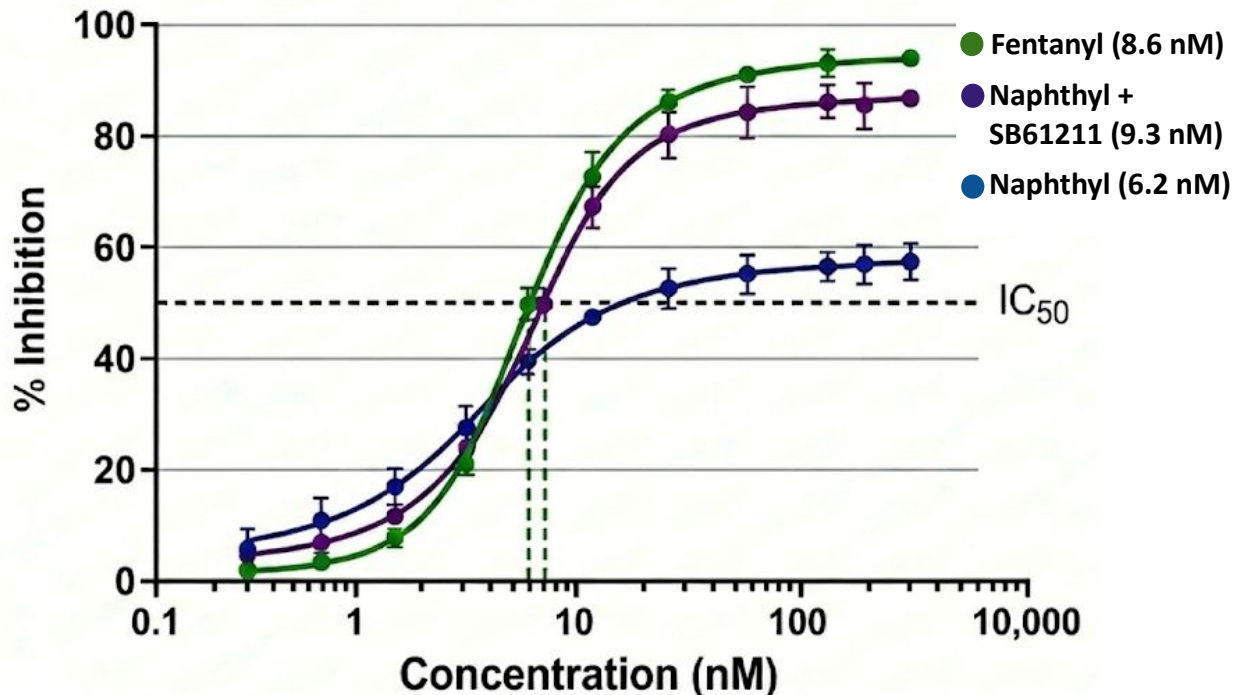


The Naphthyl-modified norfentanyl scaffold showed an IC50 of 11 nanomolar for the Nociceptin Receptor. This is superior to the Ro64-6198 of 20 nanomoles, showing superior NOP activation potency

Summary: Naphthyl-Norfentanyl compound shows potent activation of MOR and NOP receptors; Similar to high-affinity reference opioids

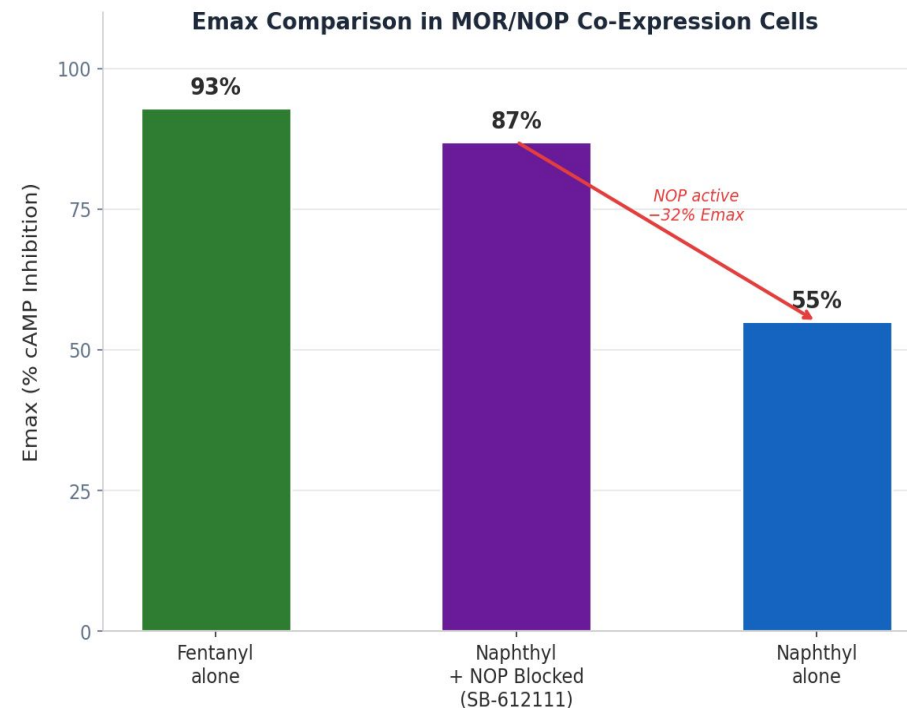
Results: Dual In-Vitro cAMP Signalling

IC50 Values - MOR/NOP Mix



In MOR/NOP co-expressing HEK293 cells, the Naphthyl scaffold showed markedly reduced cAMP suppression compared to fentanyl, demonstrating that concurrent NOP activation substantially limits the MOR-driven inhibitory ceiling

Emax Summary

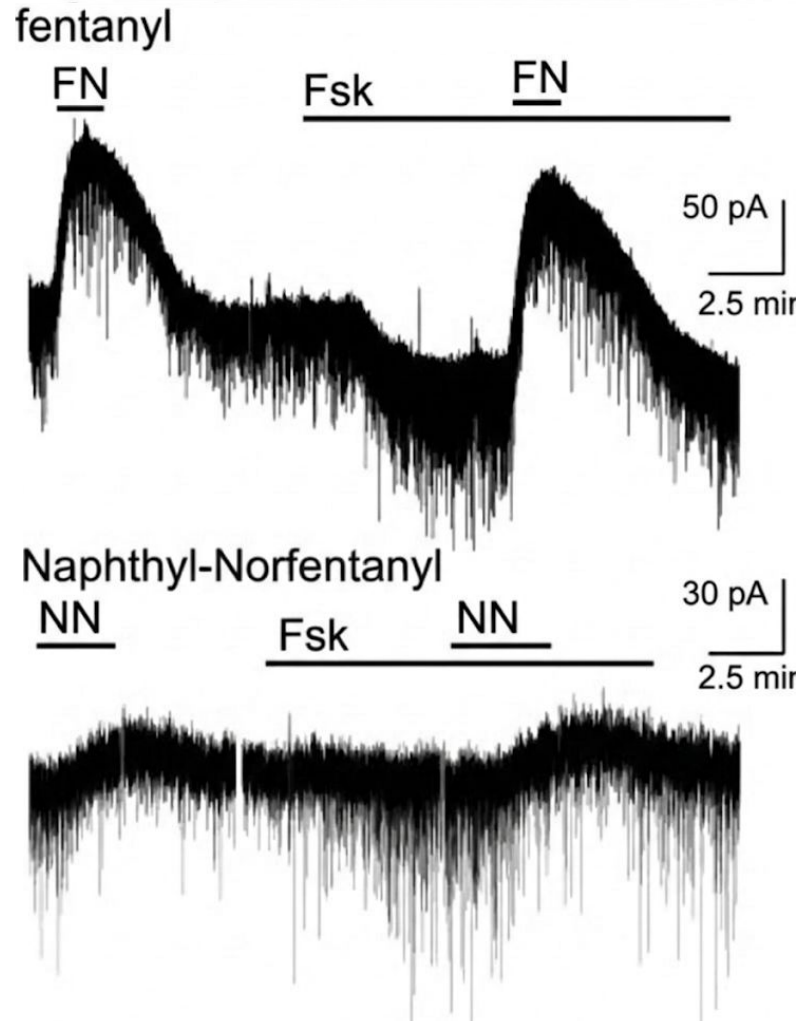


The Naphthyl-modified norfentanyl compound demonstrated an Emax of 55% cAMP inhibition in the co-expression system, compared to 93% for fentanyl— indicating that dual MOR/NOP engagement substantially limits the ceiling of receptor-driven suppression, consistent with reduced respiratory depression risk

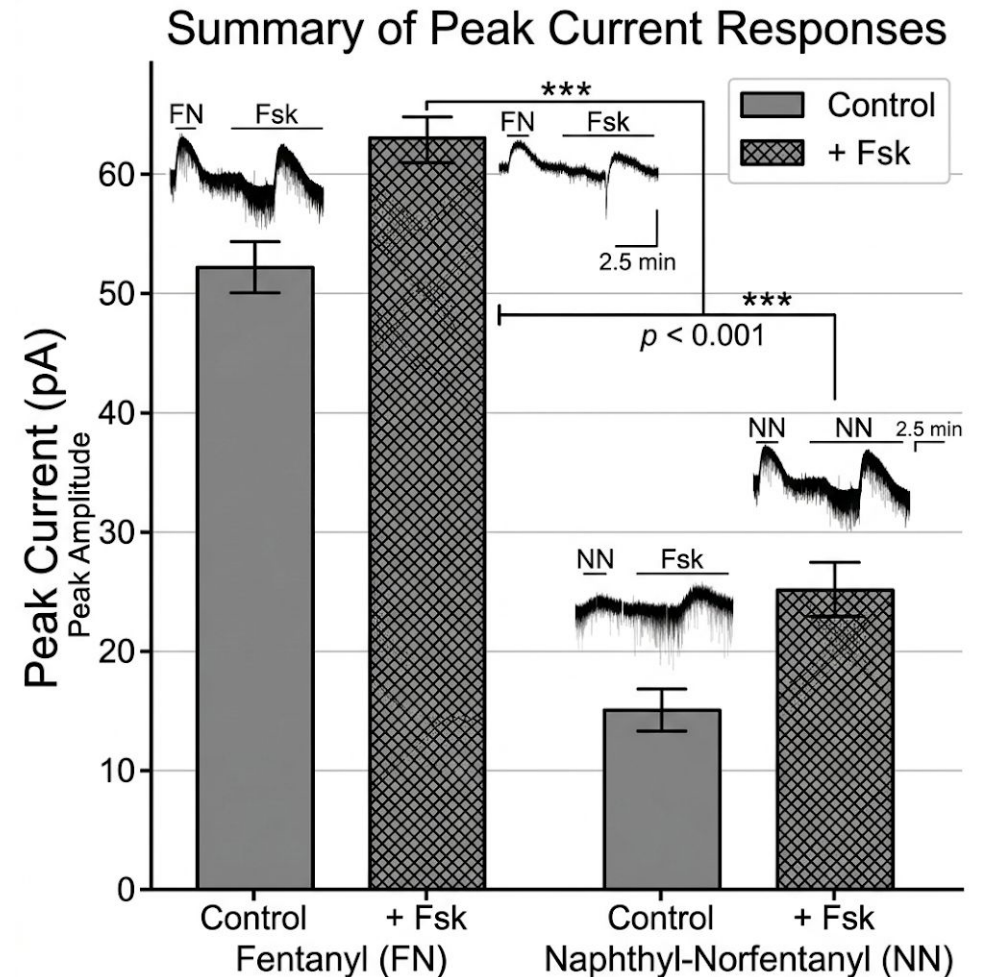
Summary: cAMP assay demonstrated that when Naphthyl-Norfentanyl engages with NOP, MOR activity is significantly reduced

Results: Electrophysiology

GIRK



GIRK Summary



Summary: GIRK Shows 2x less conductivity in mouse-derived neurons treated with Naphthyl-Norfentanyl than Fentanyl

Conclusions & Takeaways

Recap

- Fentanyl is a popular high-efficacy, fast-acting opioid. Mu-Opioid Receptor affinity induces lethal side-effects such as respiratory depression and addiction
- Identified and Synthesized Novel fentanyl-derivative with reduced side-effects
- Incorporates Naphthyl Group to target Nociceptin Receptor alongside Mu-Opioid

Highlights

- IC50 Values Comparable to Highly-potent independent agonists
- **Decreased** GIRK Conductivity (responsible for inducing respiratory depression) **by 2x**

Applications

- Enhances pain treatment as a potent, low-risk alternative to traditional opioids
- Extendable to remote locations, i.e. army, for low-risk pain management

Future Work

- Test compounds in-vitro with iPSC-derived brain cells, later in-vivo on mice
- Utilize the Naphthyl-Norfentanyl compound for potent analgesia w/heavily reduced side-effects

Key References

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